

Examining the Efficacy of Hashtag-Based Content Moderation on Instagram

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Motivation and RQs

- Content moderation on Instagram involves banning certain keywords and phrases by making them unsearchable.
- Measured the precision of Instagram’s banning approach by finding how many of these posts are actually violating Meta’s Community Guidelines.
- Conducted extensive topical analysis to evaluate the types of content penalized by Instagram’s blanket banning approach.

RQ1: How effective is Instagram’s hashtag-based blanket “banning” approach compared to a context-based moderation approach?

RQ1a: What proportion of content that uses banned hashtags violates Meta’s Community Standards?

RQ1b: How does a context-based moderation strategy perform compared to Instagram’s blanket banning approach?

RQ2: What types of content are being penalized by Instagram’s current blanket “banning” approach?

Manual Labeling

Method: Authors split up and independently labeled posts based on Instagram’s Community Guidelines (Inter-Rater Reliability = 0.92)

Results: 28 out of 500 banned posts violated Instagram’s guidelines (6% accuracy)

Implications:

- Overbanning can lead to a chilling effect hindering pro-recovery content
- Silences conversation regarding mental health rather than addressing harmful content
- Overbanning can impede users’ free speech and ability to express natural and healthy emotions on the platform (ex. exaggerated, humorous, or "edgy" expressions)

Developing a Classifier

Method: Prompted CogVLM to generate descriptions of each post; The descriptions and manual labels of 100 posts were given to train a BERT Model which classified remaining 400 posts

	Instagram			Context-Based Classifier	
Ground Truth	Banned	Not Banned	Ground Truth	Banned	Not Banned
Banned	28	0	Banned	6	16
Not Banned	472	0	Not Banned	78	300
Precision: 5.6% Recall: 100%			Precision: 7.1% Recall: 27.3%		

Implication: Classifier slightly outperformed Instagram’s current approach demonstrating potential of context-aware approaches with larger data sets

Topical Analysis

Method: One author did manual qualitative analysis on these posts. Utilized CogVLM post descriptions and performed LDA to find 7 topic clusters with high coherence.

Results: 20% of banned posts were prosocial, mental-health related content. Found 5 common post themes: Informative, Directive, Inspirational, Anecdotal, Advisory

Implication: Instagram’s current approach hinders pro-recovery and mental wellness content from reaching a wider audience

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